Section 4.7 Atthaseth Kandewutthithumrong 125010006 T03

1. a) X^2+C b) $\frac{X^3}{3}+C$ c) $\frac{X^3}{3}-X^2+X+C$

7. a) $X^{3/2}+C$ b) $\sqrt{X}+C$ c) $\frac{2}{3}X^{3/2}+2\sqrt{X}+C$

12. a) $\sin(\pi X)+C$ b) $\sin(\frac{\pi}{2}X)+C$ c) $\frac{2}{\pi}\sin(\frac{\pi}{2}X)+\pi\sin X+C$

23. $\int(\frac{1}{X^2}-X^2-\frac{1}{3})dX = -\frac{1}{X}-\frac{X^3}{3}-\frac{X}{3}+C$

32. $\int X^{-3}(X+1)\,dX = \int(X^{-2}+X^{-3})\,dX = -X^{-1}-\frac{1}{2}X^{-2}+C$

34. $\int \frac{4+\sqrt{t}}{t^3}dt = \int \frac{4}{t^3}+\frac{1}{t^{5/2}}\,dt = -\frac{2}{t^2}-\frac{2}{3t^{3/2}}+C$

52. $\int(2+\tan^2\theta)d\theta = \int(1+\sec^2\theta)\,d\theta = \theta+\tan\theta+C$

59. $\frac{d}{dx}(\frac{1}{5}\tan(5X-1)+C) = \frac{1}{5}\sec^2(5X-1)\cdot 5 = \sec^2(5X-1)$

63. a) wrong (just antiderivative for X) $(\frac{d}{dX}(\frac{X^2}{2}\sin X+C) = X\sin X+\frac{X^2}{2}\cos X)$

b) wrong (just antiderivative for sin x) $(\frac{d}{dX}(-X\cos X+C) = X\sin X-\cos X)$

c) correct $(\frac{d}{dX}(-X\cos X+\sin X+C) = X\sin X-\cos X+\cos X = X\sin X)$ (Product rule)

75. $y = 9\sqrt[3]{X}+C$

$y(-1)=-5 \rightarrow -5 = 9\sqrt[3]{-1}+C$

$C=4$ $\therefore y = 9\sqrt[3]{X}+4$

85. $\frac{dr}{dt} = -\frac{1}{t^2}+C$, $1 = \frac{dr}{dt}\Big|_{t=1} = -\frac{1}{1^2}+C_1$ $\therefore C_1 = 2 \rightarrow \frac{dr}{dt} = -\frac{1}{t^2}+2 \rightarrow r(t) = \frac{1}{t}+2t+C$

$1 = r(1) = \frac{1}{1}+2(1)+C_2 \rightarrow C_2 = -2 \rightarrow r(t) = \frac{1}{t}+2t-2$

95. $\frac{dy}{dX} = \sin X-\cos X \rightarrow y = -\cos X-\sin X+C$

$-1 = y(-\pi) = -\cos(-\pi)-\sin(-\pi)+C = 1-0+C$ $\therefore C=-2 \rightarrow y = -\cos X-\sin X-2$

99. 1) $\frac{d^2s}{dt^2} = -k \rightarrow \frac{ds}{dt} = -kt+C_1$; C_1 const

At $t=0 \rightarrow 30 = C_1$ $\therefore \frac{ds}{dt} = -kt+30 \rightarrow s = -\frac{k}{2}t^2+30t+C_2$

At $t=0 \rightarrow 0 = C_2 \rightarrow s = -\frac{k}{2}t^2+30t$

2) Set $0 = \frac{ds}{dt} = -kt+30 \rightarrow t = \frac{30}{k}$

3) At $t = \frac{30}{k}$, set $75 = s = -\frac{k}{2}(\frac{30}{k})^2+30(\frac{30}{k})$

$= -\frac{450}{k}+\frac{900}{k} = \frac{450}{k}$ $\therefore k=6$

103. From $\frac{d^2s}{dt^2} = a$, we get $\frac{ds}{dt} = at+C_1$

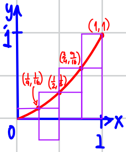
At $t=0$, $V_0 = C_1 \rightarrow \frac{ds}{dt} = at+V_0$

$\therefore s = \frac{a}{2}t^2+V_0t+C_2$

At $t=0$, $S_0 = C_2 \rightarrow S = \frac{a}{2}t^2+V_0t+S_0$ as the standard equation.

Section 5.1

1.



a) $\sum_{i=1}^{2} \min(f(c_i))\Delta X_i = 0\cdot\frac{1}{2}+\frac{1}{4}\cdot\frac{1}{2} = \frac{1}{8}$

b) $\sum_{i=1}^{4} \min(f(c_i))\Delta X_i = 0\cdot\frac{1}{4}+\frac{1}{16}\cdot\frac{1}{4}+\frac{1}{4}\cdot\frac{1}{4}+\frac{9}{16}\cdot\frac{1}{4} = \frac{7}{32}$

c) $\sum_{i=1}^{2} \max(f(c_i))\Delta X_i = \frac{1}{4}\cdot\frac{1}{2}+1\cdot\frac{1}{2} = \frac{5}{8}$

d) $\sum_{i=1}^{4} \max(f(c_i))\Delta X_i = \frac{1}{16}\cdot\frac{1}{4}+\frac{1}{4}\cdot\frac{1}{4}+\frac{9}{16}\cdot\frac{1}{4}+1\cdot\frac{1}{4} = \frac{15}{32}$

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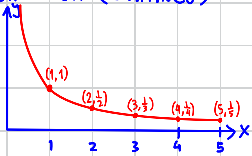
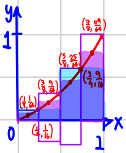
3. 

a) $\sum_{i=1}^{2} \min(f(c_i))\Delta x_i = 2\cdot\frac{1}{3} + 2\cdot\frac{1}{5} = \frac{16}{15}$

b) $\sum_{i=1}^{4} \min(f(c_i))\Delta x_i = 1\cdot\frac{1}{2} + 1\cdot\frac{1}{3} + 1\cdot\frac{1}{4} + 1\cdot\frac{1}{5} = \frac{77}{60}$

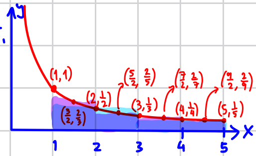
c) $\sum_{i=1}^{2} \max(f(c_i))\Delta x_i = 2\cdot 1 + 2\cdot\frac{1}{3} = \frac{8}{3}$

d) $\sum_{i=1}^{4} \max(f(c_i))\Delta x_i = 1\cdot 1 + 1\cdot\frac{1}{2} + 1\cdot\frac{1}{3} + 1\cdot\frac{1}{4} = \frac{25}{12}$

5. 

$\sum_{i=1}^{2} \text{mid}(f(c_i))\Delta x_i = \frac{1}{2}\cdot\frac{1}{16} + \frac{1}{2}\cdot\frac{9}{16} = \frac{5}{16}$

$\sum_{i=1}^{4} \text{mid}(f(c_i))\Delta x_i = \frac{1}{4}\cdot\frac{1}{64} + \frac{1}{4}\cdot\frac{9}{64} + \frac{1}{4}\cdot\frac{25}{64} + \frac{1}{4}\cdot\frac{49}{64} = \frac{21}{64}$

7. 

$\sum_{i=1}^{2} \text{mid}(f(c_i))\Delta x_i = 2\cdot\frac{1}{2} + 2\cdot\frac{1}{4} = \frac{3}{2}$

$\sum_{i=1}^{4} \text{mid}(f(c_i))\Delta x_i = 1\cdot\frac{2}{3} + 1\cdot\frac{2}{5} + 1\cdot\frac{2}{7} + 1\cdot\frac{2}{9} = \frac{496}{315}$

9. a) $\sum_{i=1}^{10} f(i-1)\cdot\Delta x_i = 0\cdot1 + 12\cdot1 + 22\cdot1 + 10\cdot1 + 5\cdot1 + 13\cdot1 + 11\cdot1 + 6\cdot1 + 2\cdot1 + 6\cdot1 = 87$ cm

b) $\sum_{i=1}^{10} f(i)\cdot\Delta x_i = 12\cdot1 + 22\cdot1 + 10\cdot1 + 5\cdot1 + 13\cdot1 + 11\cdot1 + 6\cdot1 + 2\cdot1 + 6\cdot1 + 0\cdot1 = 87$ cm

19. a) Lower: $\sum_{i=1}^{5} \min(f(x_i))\cdot\Delta x_i = 1\cdot50 + 1\cdot70 + 1\cdot97 + 1\cdot136 + 1\cdot190 = 543$ L

Upper: $\sum_{i=1}^{5} \max(f(x_i))\cdot\Delta x_i = 1\cdot70 + 1\cdot97 + 1\cdot136 + 1\cdot190 + 1\cdot265 = 758$ L

b) Lower: $\sum_{i=1}^{8} \min(f(x_i))\cdot\Delta x_i = 1\cdot50 + 1\cdot70 + 1\cdot97 + 1\cdot136 + 1\cdot190 + 1\cdot265 + 1\cdot369 + 1\cdot516 = 1693$ L

Upper: $\sum_{i=1}^{8} \max(f(x_i))\cdot\Delta x_i = 1\cdot70 + 1\cdot97 + 1\cdot136 + 1\cdot190 + 1\cdot265 + 1\cdot369 + 1\cdot516 + 1\cdot720 = 2363$ L

c) Worst case: After 8 h $\rightarrow$ oil left $= 25000 - 2363 = 22637$ L

Time taken $= \frac{22637}{720} = 31.44$ h

Best case: After 8 h $\rightarrow$ oil left $= 25000 - 1693 = 23307$ L

Time taken $= \frac{23307}{720} = 32.37$ h

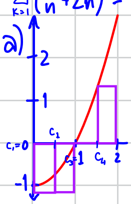
Section 5.2

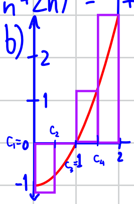
8. a) $\sum_{k=1}^{6} 2^{k-1} = 1 - 2 + 4 - 8 + 16 - 32$

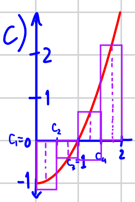
12. $\sum_{k=1}^{4} k^2$

30. $\sum_{k=9}^{36} k = 9 + 10 + 11 + \dots + 35 + 36 = \frac{(9+36)\times 28}{2} = 630$

32. $\sum_{k=1}^{n} \left(\frac{1}{n} + 2n\right) = n\left(\frac{1}{n} + 2n\right) = 1 + 2n^2$

33. a) 

b) 

c) 

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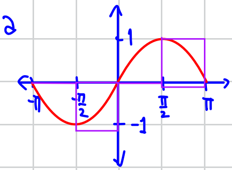
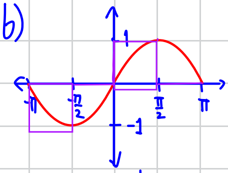
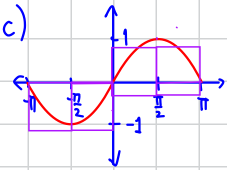
35. a



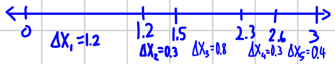
b)



c)

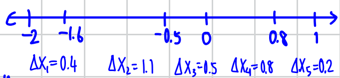


37.



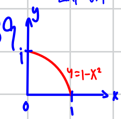
$\therefore \|P\| = \max(\Delta X_1, \Delta X_2, \Delta X_3, \Delta X_4, \Delta X_5) = 1.2$

38.



$\therefore \|P\| = \max(\Delta X_1, \Delta X_2, \Delta X_3, \Delta X_4, \Delta X_5) = 1.1$

39.

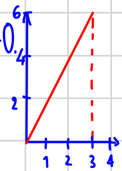


$f(c_i) = 1 - X_i^2 = 1 - \left(\frac{i}{n}\right)^2, \ \Delta X_i = \frac{1}{n},$

$\therefore \sum_{i=1}^{n} f(c_i)\Delta X_i = \sum_{i=1}^{n}\left(\frac{1}{n} - \frac{i^2}{n^3}\right) = \frac{1}{n}\cdot n - \frac{n(n+1)(2n+1)}{6n^3} = 1 - \frac{(n+1)(2n+1)}{6n^2}$

$\lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\Delta X_i = \lim_{n\to\infty} 1 - \frac{n(1+\frac{1}{n})\cdot n(2+\frac{1}{n})}{6n^2} = 1 - \frac{1\cdot 2}{6} = \frac{2}{3}$

40.



$f(c_i) = 2X_i = 2\left(\frac{3i}{n}\right), \ \Delta X_i = \frac{3}{n} \rightarrow X_i = \frac{3i}{n}$

$\therefore \sum_{i=1}^{n} f(c_i)\Delta X_i = \sum_{i=1}^{n}\left(\frac{6i}{n}\right)\left(\frac{3}{n}\right) = \sum_{i=1}^{n}\frac{18i}{n^2} = \frac{18}{n^2}\left(\frac{n(n+1)}{2}\right) = 9\left(\frac{n+1}{n}\right)$

$\therefore \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\Delta X_i = 9(1) = 9$

Section 5.3

1. $\int_0^2 x^2\,dx$

3. $\int_{-7}^{5} (x^2 - 3x)\,dx$

6. $\int_0^1 \sqrt{4-x^2}\,dx$

8. $\int_0^{\pi/4} \tan x\,dx$

9. a) $\int_2^2 g(x)\,dx = 0$

b) $\int_5^1 g(x)\,dx = -\int_1^5 g(x)\,dx = -8$

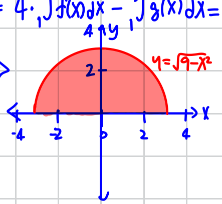
c) $\int_1^2 3f(x)\,dx = 3\int_1^2 f(x)\,dx = 3(-4) = -12$

d) $\int_2^5 f(x)\,dx = \int_1^5 f(x)\,dx - \int_1^2 f(x)\,dx = 6 - (-4) = 10$

e) $\int_1^5 [f(x) - g(x)]\,dx = \int_1^5 f(x)\,dx - \int_1^5 g(x)\,dx = 6 - 8 = -2$

f) $\int_1^5 [4f(x) - g(x)]\,dx = 4\cdot\int_1^5 f(x)\,dx - \int_1^5 g(x)\,dx = 4(6) - 8 = 16$

17. $\int_{-3}^{3} \sqrt{9-x^2}\,dx \Rightarrow$



Area under graph $= \frac{1}{2}\pi(3^2) = \frac{9\pi}{2}$

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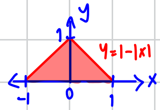
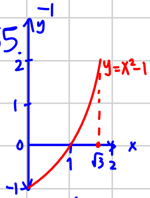
20. $\int_{-1}^{1}(1-|x|)\,dx =$



Area under graph $= \frac{1}{2}\cdot 2\cdot 1 = 1$

55.



Average value $= \frac{1}{\sqrt{3}-0}\int_0^{\sqrt{3}}(x^2-1)\,dx = \frac{1}{\sqrt{3}}\left(\frac{x^3}{3}-x\right)\Big|_0^{\sqrt{3}} = \frac{1}{\sqrt{3}}\left(\frac{3\sqrt{3}}{3}-\sqrt{3}\right) = 0$

71. $x-x^2$ is positive when $0<x<1$ (Other values of x will make the value negative)
Hence, $\int_a^b (x-x^2)\,dx$ will be maximized when $a=0$ and $b=1$.

73. Since $\frac{1}{1+x^2} \geq \frac{1}{1+1^2} = \frac{1}{2}$ and $\frac{1}{1+x^2} \leq \frac{1}{1+0^2} = 1$ for all $x\in[0,1]$
Hence, $\int_0^1 \frac{1}{1+x^2}\,dx \geq \int_0^1 \frac{1}{2}\,dx = \frac{1}{2}$ (Lower bound) and $\int_0^1 \frac{1}{1+x^2}\,dx \leq \int_0^1 1\,dx = 1$ (Upper bound)

76. Since for all $x\in[0,1]$, $\sqrt{x+8} > \sqrt{0+8} = \sqrt{8}$ and $\sqrt{x+8} \leq \sqrt{1+8} = 3$
So, $\int_0^1 \sqrt{x+8}\,dx \geq \int_0^1 \sqrt{8}\,dx = \sqrt{8} = 2\sqrt{2}$ and $\int_0^1 \sqrt{x+8}\,dx \leq \int_0^1 3\,dx \quad 3$

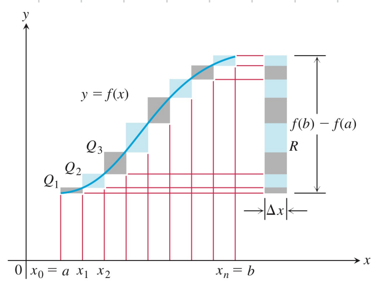
Hence, the value of $\int_0^1 \sqrt{x+8}\,dx$ is between $2\sqrt{2}$ and 3.

77. Since $f(x)\geq 0$ on $[a,b]$, we get $\int_a^b f(x)\,dx \geq \int_a^b 0\,dx = 0$ by Max-min inequality.

78. Since $f(x)\leq 0$ on $[a,b]$, we get $\int_a^b f(x)\,dx \leq \int_a^b 0\,dx = 0$ by Max-min inequality

81. $\int_a^b av(f)\,dx = \int_a^b \underbrace{\left(\frac{1}{b-a}\int_a^b f(x)\,dx\right)}_{\text{Constant } K} dx = (b-a)\left(\frac{1}{b-a}\int_a^b f(x)\,dx\right) = \int_a^b f(x)\,dx \rightarrow$ Yes, it is.

83. a)



From the figure, since f rises steadily as x moves from left to right, we get left sum as the lower sum and right sum as the upper sum.
Lower sum $= f(a)\cdot\Delta x + f(x_1)\Delta x + \ldots + f(x_{n-1})\Delta x$ — ①
Upper sum $= f(x_1)\cdot\Delta x + f(x_2)\Delta x + \ldots + f(b)\cdot\Delta x$ — ②
② − ① Difference between upper sum and lower sum
$= (f(b)-f(a))\cdot\Delta x$

b) $L = f(a)\cdot\Delta x_1 + f(x)\cdot\Delta x_2 + \ldots + f(x_{n-1})\cdot\Delta x_n$ — ①
$U = f(x_1)\cdot\Delta x_1 + f(x_2)\cdot\Delta x_2 + \ldots + f(b)\cdot\Delta x_n$ — ②

② − ① : $U - L = (f(x_1)-f(a))\cdot\Delta x_1 + (f(x_2)-f(x_1))\cdot\Delta x_2 + \ldots + (f(b)-f(x_{n-1}))\cdot\Delta x_n$
$\leq (f(x_1)-f(a))\cdot\Delta x_{max} + (f(x_2)-f(x_1))\cdot\Delta x_{max} + \ldots + (f(b)-f(x_{n-1}))\cdot\Delta x_{max}$
$= \Delta x_{max}\cdot(f(b)-f(a))$

$\therefore \lim_{\|P\|\to 0} U-L = \lim_{\|P\|\to 0} \Delta x_{max}\cdot(f(b)-f(a)) = \lim_{\|P\|\to 0}(\|P\|\cdot(f(b)-f(a))) = 0$